



Indiana Michigan Power
Cook Nuclear Plant
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Bridgman, MI 49106
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September 28, 2023

AEP-NRC-2023-43
10 CFR 50.73

Docket No.: 50-315

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
11555 Rockville Pike
Rockville, MD 20852

Donald C. Cook Nuclear Plant Unit 1
LICENSEE EVENT REPORT 315/2023-001-00
Two Trains of Fuel Handling Area Exhaust Ventilation Inoperable

In accordance with 10 CFR 50.73, Licensee Event Report (LER) System, Indiana Michigan Power Company, the licensee for Donald C. Cook Nuclear Plant Unit 1, is submitting as an enclosure to this letter the following report:

LER 315/2023-001-00: Two Trains of Fuel Handling Area Exhaust Ventilation Inoperable

There are no commitments contained in this submittal.

Should you have any questions, please contact Mr. Michael K. Scarpello, Regulatory Affairs Director, at (269) 466-2649.

Sincerely,

Kelly J Ferneau
Site Vice President

SJM/sjh

Enclosure: Licensee Event Report 315/2023-001-00: Two Trains of Fuel Handling Area Exhaust Ventilation Inoperable

c: EGLE – RMD/RPS
J. B. Giessner – NRC Region III
NRC Resident Inspector
N. Quilico - MPSC
R. M. Sistevaris – AEP Ft. Wayne
S. P. Wall – NRC, Washington D.C.
A. J. Williamson – AEP Ft. Wayne

Enclosure to AEP-NRC-2023-43

Licensee Event Report 315/2023-001-00: Two Trains of Fuel Handling Area Exhaust
Ventilation Inoperable



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-m/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to infocollections@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Donald C. Cook Nuclear Plant Unit 1

☒ 050☐ 052

2. Docket Number

00315

3. Page

1 OF 4

4. Title

Two Trains of Fuel Handling Area Exhaust Ventilation Inoperable

5. Event Date

6. LER Number

7. Report Date

8. Other Facilities Involved

Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
07	31	2023	2023	- 01 -	00	09	28	2023	Donald C. Cook Unit 2		00316
Facility Name										☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☒ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Michael K. Scarbello, Regulatory Affairs Director

Phone Number (Include area code)

269-466-2649

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	VL	UDMP	Ruskin	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month Day Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On July 25, 2023, with Donald C. Cook Nuclear Plant Units 1 and 2 operating at 100 percent power, Operators identified the Fuel Handling Area Exhaust Ventilation (FHAEV) System filter unit difference in pressure (DP) readings were below administrative limits. Subsequently, irradiated fuel movement occurred in the spent fuel pool on July 31, 2023 and August 1, 2023. Later inspection of each train FHAEV System backdraft damper on August 2, 2023 identified degraded conditions that rendered each FHAEV train inoperable back to July 23, 2023.

Technical Specifications require one train of the FHAEV System to be operable during movement of irradiated fuel assemblies in the auxiliary building. There were two occurrences of irradiated fuel movement in the auxiliary building during the time period when both FHAEV trains were inoperable.

The event is reportable per 10 CFR 50.73(a)(2)(i)(B) "Operation or Condition Prohibited by Technical Specifications", 10 CFR 50.73(a)(2)(v)(D) "Event or Condition that Could Have Prevented Fulfillment of a Safety Function", and 10 CFR 50.73(a)(2)(vii) "Common Cause Inoperability of Independent Trains or Channels".

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Donald C. Cook Nuclear Plant Unit 1	☒ 050	2. DOCKET NUMBER 00315	YEAR 2023	3. LER NUMBER SEQUENTIAL NUMBER 001	REV NO. 00
	☐ 052				

NARRATIVE**Event Description**

On July 25, 2023, the Donald C. Cook Nuclear Plant Units 1 and 2 were operating at 100 percent power.

Operators identified the Fuel Handling Area Exhaust Ventilation (FHAEV) System [VL] high efficiency particulate air filter (HEPA) and activated charcoal adsorber section difference in pressure (DP) readings were below administrative limits and commenced troubleshooting activities.

Operators observed later during a system walkdown that with 12-HV-AFX-2, "Auxiliary Building Ventilation Fuel Handling Area Exhaust Fan #2" [VL][FAN] operating, there was reverse flow sufficient to cause "windmilling" of the opposite train fan 12-HV-AFX-1 "Auxiliary Building Ventilation Fuel Handling Area Exhaust Fan #1" [VL][FAN] while in standby. Similar symptoms of reverse flow were observed on 12-HV-AFX-2 when idle, while 12-HV-AFX-1 was operating.

Internal inspections of each train FHAEV System backdraft damper assemblies 12-HV-BD-AFX-1, "Auxiliary Building Ventilation Fuel Handling Area Exhaust Fan HV-AFX-1 Discharge Backdraft Damper" [VL][UDMP] and 12-HV-BD-AFX-2, "Auxiliary Building Ventilation Fuel Handling Area Exhaust Fan HV-AFX-2 Discharge Backdraft Damper" [VL][UDMP] identified both damper mechanisms were partially stuck open and degraded which caused reduced system airflow. The identified conditions rendered each FHAEV System train inoperable. An engineering review of air flow data from 1-VFR-315 "Unit Vent Stack Flow Recorder Transmitter" [VL][FT] indicated system air flow was less than the operability range following a swap from 12-HV-AFX-1 to 12-HV-AFX-2, on July 23, 2023. No symptoms of degraded conditions were detected prior to that time.

Repairs to the backdraft dampers were completed and required surveillance testing restored the FHAEV System to an operable status on August 5, 2023.

Subsequently, during the time period between July 23, 2023 and August 5, 2023, when the FHAEV System was inoperable, irradiated fuel was moved in the auxiliary building on July 31, 2023 and August 1, 2023.

Technical Specification Limiting Condition for Operation (LCO) 3.7.13 "Fuel Handling Area Exhaust Ventilation (FHAEV) System" requires "One FHAEV train shall be OPERABLE and in operation" and is applicable "During movement of irradiated fuel assemblies in the auxiliary building."

LCO 3.7.13 requires the FHAEV system to be capable of removing airborne activity originating from a failed fuel assembly any time irradiated fuel is being moved. The mission time of the system is 30 days, which includes the radiological consequences analysis of record for a fuel handling accident in the auxiliary building, which credits the FHAEV System to filter the release through the unit vent. This filtered pathway is credited for the entire analyzed release duration. In order to accomplish the filtration safety function, the Bases for LCO 3.7.13 specifies for each FHAEV train, that ductwork, valves, and dampers are operable, and air flow can be maintained.

Therefore, the event is reportable under 10 CFR 50.73(a)(2)(i)(B) "Operation or Condition Prohibited by Technical Specifications" due to irradiated fuel being moved when the FHAEV System was incapable of performing its safety function, contrary to LCO 3.7.13 licensing basis requirements; 10 CFR 50.73(a)(2)(v)(D) "Event or Condition that Could Have Prevented Fulfillment of a Safety Function" due to neither train of the FHAEV being capable of performing the filtration safety function during movement of irradiated fuel; and, 10 CFR 50.73(a)(2)(vii) "Common Cause Inoperability of Independent Trains or Channels", when two independent trains (backdraft dampers) of the single system (FHAEV) were inoperable and would have impacted the ability to mitigate the consequences of an accident.

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NARRATIVE**Component**

12-HV-BD-AFX-1, "Auxiliary Building Ventilation Fuel Handling Area Exhaust Fan HV-AFX-1 Discharge Backdraft Damper"; Ruskin Type CBS4.

12-HV-BD-AFX-2, "Auxiliary Building Ventilation Fuel Handling Area Exhaust Fan HV-AFX-2 Discharge Backdraft Damper"; Ruskin Type CBS4.

Cause of the Event

The direct cause of the event was degradation and failure of each train FHAEV backdraft damper that lowered air flow rates below system operability limits due to recirculation air flow through the opposite train.

A causal evaluation is in progress at the time of this report. If substantial information is identified that would significantly change the perception of the event or planned corrective actions, a supplemental Licensee Event Report will be submitted at that time.

Assessment of Safety Consequences**Nuclear Safety**

There were no actual nuclear safety hazards that resulted from movement of irradiated fuel in the auxiliary building with the FHAEV system inoperable.

Potential nuclear safety hazards were evaluated for a 30 day mission time regarding radiological dose consequences of a postulated Fuel Handling Accident (FHA) occurring in the auxiliary building at the time of the event given the as-found conditions. When crediting the delay time between the last reactor shutdown for refueling (Fall 2022) and when the FHAEV system was inoperable, the current D. C. Cook Unit 1 and Unit 2 radiological dose consequences analysis of record for the FHA in the Updated Final Safety Analysis Report (UFSAR) bounds the plant configuration. Therefore, the safety significance is very low.

Industrial Safety

There were no actual personnel safety hazards that resulted from movement of irradiated fuel in the auxiliary building with the FHAEV system inoperable.

Radiological Safety

There were no actual radiological safety hazards or radiological releases that resulted from movement of irradiated fuel in the auxiliary building with the FHAEV system inoperable.

Completed Actions

1. Repairs of the backdraft dampers and surveillance testing were completed to restore the FHAEV system to an operable status.

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1. FACILITY NAME

050



052

Donald C. Cook Nuclear Plant Unit 1

2. DOCKET NUMBER

00315

3. LER NUMBER

YEAR

SEQUENTIAL
NUMBERREV
NO.

2023

001

00

NARRATIVE**Planned Corrective Actions**

1. Evaluate and revise the preventive maintenance strategy for the backdraft dampers.
2. Revise procedures to add FHAEV system functional checks to verify appropriate positioning of the backdraft dampers prior to movement of fuel within the storage pool or crane operations with loads over the storage pool.

Previous Similar Events

A review of LERs for the past five years involving similar events was performed. None of the events occurred from similar causes.